	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2025 – 2026*

Presentation on Capstone Project Execution (CPE–316004)

Topic	Co Attainment Calculation For Outcome Based Education	
Presented by	1.Ruchika Dilip Patil.	
	2.Krishna VinodKumar Popali.	
	3.Aman Anjum Iqbal Shaikh.	
	4.Suvarna Mahendra Sonawane.	
Guided by	Ms. Chaitali S. Patil	
Semester	**Sixth (K Scheme)**	**Date ://2026**

Abstract

This project presents a system that will support continuous academic evaluation during the ongoing teaching process. The system will enable teachers to submit assessment data through a centralized platform, where performance levels will be processed using predefined rubrics. Based on this evaluation, the system will identify attainment status and highlight the gap between expected and achieved outcomes. The generated results will be presented in a structured report format, allowing timely review within the same academic Assessment. By providing immediate analytical feedback, the system will support informed academic decisions and will help faculty focus on improving instructional effectiveness rather than manual data processing.

Content

1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

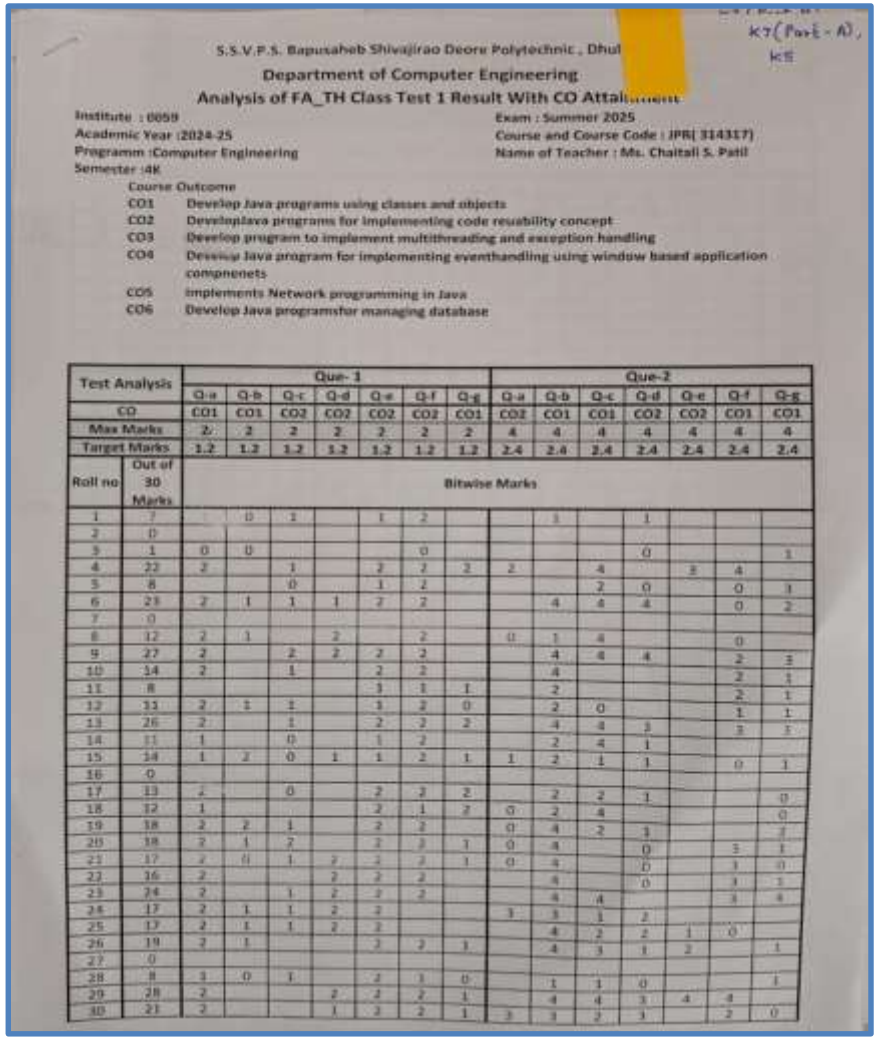
1. Introduction

In academic institutions, the evaluation of student performance requires systematic analysis to understand learning effectiveness. Traditionally, this evaluation process involved maintaining student assessment records separately and performing calculations after completion of assessments. Such practices limited the ability to observe performance trends during the teaching period. Since the evaluation was carried out at a later stage, instructors had minimal opportunity to adjust teaching strategies for the same group of students. Additionally, the dependence on manual record handling increased processing time and reduced efficiency. These challenges highlight the importance of adopting structured mechanisms that support timely performance analysis and enable instructional refinement within the academic term.

2. Literature Review

Earlier studies show that many institutions calculate attainment levels using spreadsheet-based methods. In this process, teachers manually enter student marks and later calculate attainment values and performance gaps. This calculation is usually done after completing all assessments of the course. Because of this delay, teachers cannot explain the results or the identified gaps to the current batch of students during the same academic term. The analysis becomes useful mainly for the next batch, where teachers can plan assignments or improvements based on the previously identified gaps. However, the current batch does not get immediate feedback about their performance or the areas where improvement is required. This limitation highlights the need for a system that can provide timely analysis and support better academic monitoring.

2. Literature Review



S.S.V.P.S. Bapusaheb Shivajirao Deore Polytechnic , Dhul

Department of Computer Engineering

Analysis of FA_TH Class Test 1 Result With CO Attai...

Institute : 0059
Academic Year :2024-25
Programm :Computer Engineering
Semester :4K

Exam : Summer 2025
Course and Course Code : JPR(314317)
Name of Teacher : Ms. Chaitali S. Patil

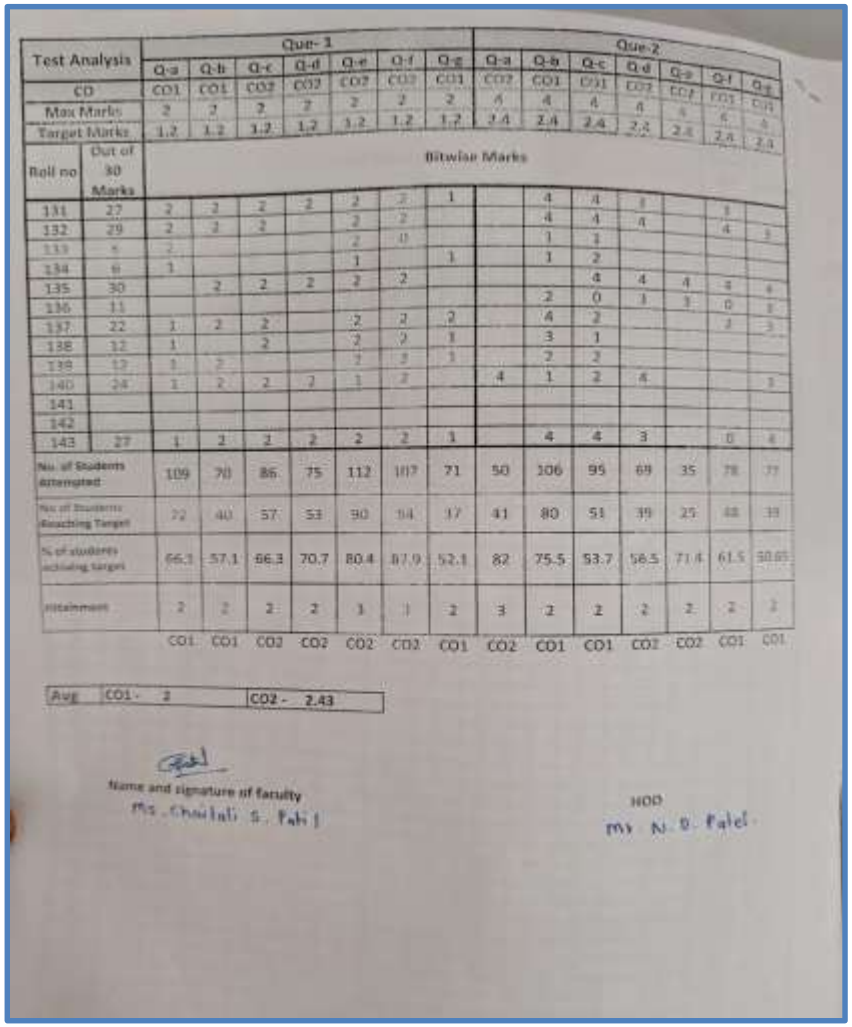
Course Outcome

- CO1 Develop Java programs using classes and objects
- CO2 Developjava programs for implementing code reusability concept
- CO3 Develop program to implement multithreading and exception handling
- CO4 Develop Java program for implementing eventhandling using window based application componenets
- CO5 Implements Network programming in Java
- CO6 Develop Java programsfor managing database

Test Analysis		Que- 1							Que-2						
		Q-a	Q-b	Q-c	Q-d	Q-e	Q-f	Q-g	Q-a	Q-b	Q-c	Q-d	Q-e	Q-f	Q-g
CO		CO1	CO1	CO2	CO2	CO2	CO2	CO1	CO2	CO1	CO1	CO2	CO2	CO1	CO1
Max Marks		2	2	2	2	2	2	2	4	4	4	4	4	4	4
Target Marks		1.2	1.2	1.2	1.2	1.2	1.2	1.2	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Roll no	Out of 30 Marks	Bitwise Marks													
1	7		0	1		1	2			1		1			
2	0														
3	1	0	0				0					0			1
4	22	2		1		2	2	2	2		4		3	4	
5	8			0		1	2				2	0		0	1
6	23	2	1	1	1	2	2			4	4	4		0	2
7	0														
8	12	2	1		2		2		0	1	4			0	
9	27	2		2	2	2	2			4	4	4		2	3
10	14	2		1		2	2			4				2	1
11	8					1	1	1		2				2	1
12	31	2	1	1		1	2	0		2	0			1	1
13	26	2		1		2	2	2		4	4	3		3	3
14	11	1		0		1	2			2	4	1			
15	14	1	2	0	1	1	2	1	1	2	1	1		0	1
16	0														
17	13	2		0		2	2	2		2	2	1			0
18	12	1				2	1	2	0	2	4				0
19	18	2	2	1		2	2		0	4	2	3			2
20	18	2	1	2		2	2	1	0	4		0		3	1
21	17	2	0	1	2	2	2	1	0	4		0		3	0
22	16	2			2	2	2			4		0		3	3
23	24	2		1	2	2	2			4	4			3	4
24	17	2	1	1	2	2			3	3	3	2			
25	17	2	1	1	2	2				4	2	2	1	0	
26	19	2	1			2	2	1		4	3	1	2		1
27	0														
28	8	1	0	1		2	1	0		1	1	0			1
29	28	2			2	2	2	1		4	4	3	4	4	
30	21	2			1	2	2	1	3	3	2	3		2	0

Fig. Bitwise Class Test 1 Calculation

2. Literature Review



Test Analysis		Que- 1							Que-2						
		Q-a	Q-b	Q-c	Q-d	Q-e	Q-f	Q-g	Q-a	Q-b	Q-c	Q-d	Q-e	Q-f	Q-g
CO		CO1	CO1	CO2	CO2	CO2	CO2	CO1	CO2	CO1	CO1	CO2	CO2	CO1	CO1
Max Marks		2	2	2	2	2	2	2	4	4	4	4	4	4	4
Target Marks		1.2	1.2	1.2	1.2	1.2	1.2	1.2	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Roll no	Out of 30 Marks	Bitwise Marks													
131	27	2	2	2	2	2	2	1		4	4	3		3	
132	29	2	2	2		2	2			4	4	4		4	3
133	5	2				2	0			1	1				
134	6	1				1		1		1	2				
135	30		2	2	2	2	2				4	4	4	4	4
136	11									2	0	3	3	0	3
137	22	1	2	2		2	2	2		4	2			3	3
138	12	1		2		2	2	1		3	1				
139	12	1	2			2	3	1		2	2				
140	24	1	2	2	2	1	2		4	1	2	4			3
141															
142															
143	27	1	2	2	2	2	2	1		4	4	3		0	4
No. of Students attempted		109	70	86	75	112	107	71	50	106	95	69	35	78	77
No. of Students Reaching Target		72	40	57	53	90	94	37	41	80	51	39	25	48	39
% of students achieving target		66.1	57.1	66.3	70.7	80.4	87.9	52.1	82	75.5	53.7	56.5	71.4	61.5	50.65
Attainment		2	2	2	2	1	1	2	3	2	2	2	2	2	2
		CO1	CO1	CO2	CO2	CO2	CO2	CO1	CO2	CO1	CO1	CO2	CO2	CO1	CO1

Avg	CO1 - 2	CO2 - 2.43

Name and signature of faculty
Ms. Chaitali S. Patil

HOD
Mr. N. D. Patel

Fig. Bitwise Class Test 1 Calculation

2. Literature Review

S.S.V.P.S. Bapusaheb Shivajirao Deore Polytechnic , Dhule
Department of Computer Engineering
SELF LEARNING ASSESSMENT (SLA)
Assignment CO Attainment

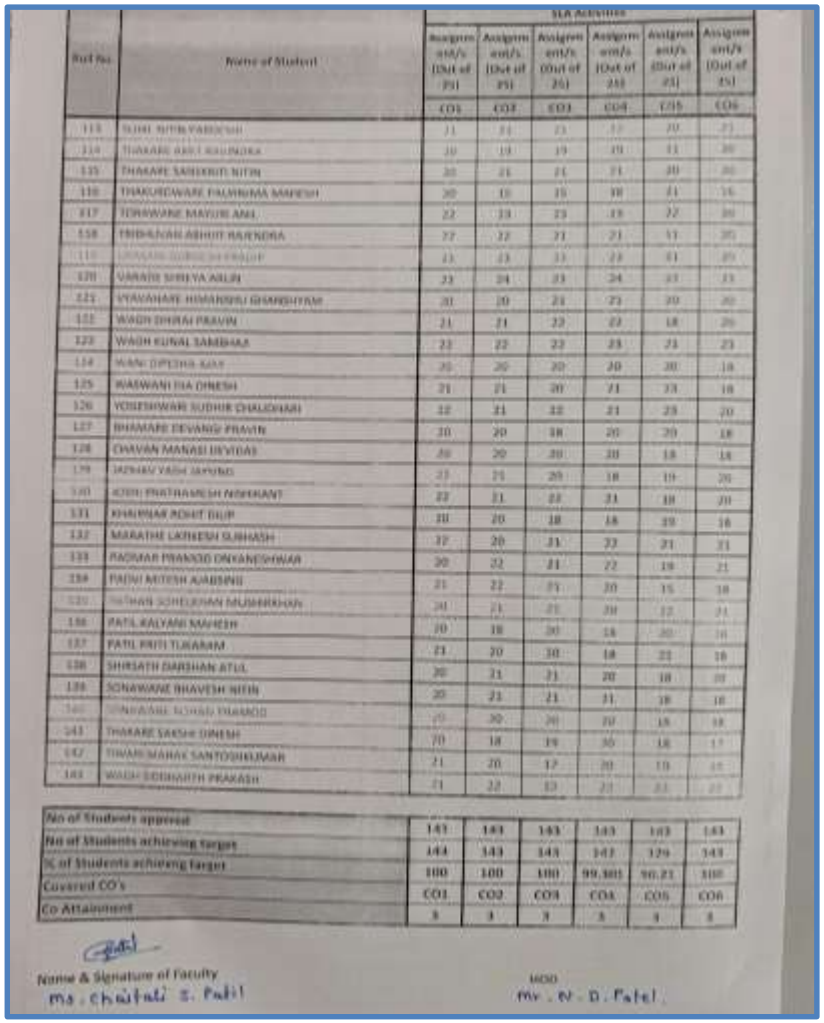
Institute Code and Name: 0059
Academic Year : 2024-25
Program:CO4K
Semester: IV

Exam: Summer 2025
Course: JPR Course Code : 314317
Marks: Max- 25 Min - 10
Name of Faculty : Ms. Chaitali S. Patil

Roll No.	Name of Student	SLA Activities					
		Assignment/s (Out of 25)	Assignment/s (Out of 25)	Assignment/s (Out of 25)	Assignment/s (Out of 25)	Assignment/s (Out of 25)	Assignment/s (Out of 25)
		CO1	CO2	CO3	CO4	CO5	CO6
1	AHIRE SIDDHESH SUNIL	20	17	18	19	16	17
2	AHIRE TANUSHREE PARAG	18	17	15	14	14	16
3	ANSARI MOHAMMAD SOHAIL MOHAMMAD HUSAIN	15	16	17	16	14	16
4	ANSARI ZEBA JAVEED IQBAL	21	22	21	20	21	18
5	BAISANE NARENDRA PRABHAKAR	22	23	24	22	16	20
6	BHAMARE PIYUSH PRASHANT	22	24	23	23	21	24
7	BHAMARE VAISHNAVI SUNIL	18	16	17	20	14	18
8	BHAVSAR SATYAM PARESH	22	23	23	23	16	20
9	BOHRI HATIM HUZEFA	16	17	19	18	21	18
10	BORASE PRATHAMESH UDAY	19	18	20	19	16	19
11	BORSE AASHUTOSH KANHAIYALAL	21	22	23	22	16	18
12	BUVA SAKSHI ANANDGIR	21	21	19	19	20	18
13	CHAUDHARI KARTIK BHAGWAN	18	18	18	18	21	18
14	CHAUDHARI KHUSHI PRAVIN	21	22	22	22	20	19
15	CHAUDHARI KUNAL RAGHUNATH	19	18	18	20	18	18
16	CHAUDHARI SAKSHI JAIPRAKASH	18	20	19	19	11	18
17	CHAUDHARI SIDDHESH PRAKASH	20	20	18	16	17	18
18	CHAUDHARI YASHIKA PARESH	17	16	15	16	16	16
19	CHAVAN DIKSHA DAGADU	20	21	20	20	22	20
20	CHAVAN PRADNYA VALMIK	17	17	19	18	22	18
21	CHAVAN SAKSHI VINOD	22	21	22	21	22	20
22	CHETAN JAGDISH PATEL	22	21	21	21	18	21
23	CHUGER HANISH SUNIL	19	18	17	17	18	18
24	DEORE DURGESH TRYAMBAK	16	20	19	17	18	18
25	DEORE KISHORI NITIN	17	17	19	19	19	18
26	DEORE MOHAN NAVANEET	20	15	15	18	20	16
27	DEORE REVATI NITIN	22	21	21	22	11	20
28	DEORE SNEHAL TUKARAM	20	18	18	19	20	18
29	DERE AMEY PRAVIN	22	23	23	22	22	23
30	DESALE BHUMIKA SUDHIR	21	20	20	21	22	20

Fig.(SLA)Self Learning Assessment

2. Literature Review



Roll No.	Name of Student	SLA Activities					
		Assignment/s (Out of 25)	Assignment/s (Out of 25)	Assignment/s (Out of 25)	Assignment/s (Out of 25)	Assignment/s (Out of 25)	Assignment/s (Out of 25)
		CO1	CO2	CO3	CO4	CO5	CO6
113	SUNI NITIN PARDESHI	21	21	21	22	20	21
114	THAKARE ABHI KAUNDRA	20	19	19	19	21	20
115	THAKARE SANSKRUTI NITIN	20	21	21	21	20	20
116	THAKURDWARE PALWISHA MAHESH	20	19	19	18	21	16
117	TONAWANE MAYURI ANIL	22	19	23	19	22	20
118	TRIBHUVAN ABHIJIT RAVINDRA	22	22	21	21	21	20
119	[illegible]	21	21	21	21	21	20
120	VARADE SHREYA ARUN	23	24	23	24	23	23
121	VYAVAHARE HIMANSHU GHANSHYAM	20	20	21	21	20	20
122	WAGH DHIRAJ PRAVIN	21	21	22	22	18	20
123	WAGH KUNAL SAMBHAJI	22	22	22	23	21	23
124	WANI DIPESHA AJAY	20	20	20	20	20	18
125	WAIWANI JIA DINESH	21	21	20	21	23	18
126	YOSEPHWARE SUDHIR CHALCHARI	22	21	22	21	23	20
127	BHAMARE DEVANSH PRAVIN	20	20	18	20	20	18
128	CHAVAN MANASI DEVIDAS	20	20	20	20	18	18
129	JADHAV VAISHNAVI [illegible]	23	21	20	18	19	20
130	KOTRI PRATHAMESH NISHANT	22	21	22	21	18	20
131	KHAIRNAR ROHIT DILIP	20	20	18	18	19	18
132	MARATHE LATIKESH SUBHASH	22	20	21	22	21	21
133	PACHAR PRAMOD DNYANESHWAR	20	22	21	22	19	21
134	PADVI MITESH AJABSING	21	22	21	20	15	18
135	PATHAN SOHELKHAN MUSHIRKHAN	20	21	21	20	12	21
136	PATIL KALYANI MAHESH	20	18	20	18	20	16
137	PATIL PRITI TUKARAM	21	20	20	18	22	18
138	SHIRSATH DARSHAN ATUL	20	21	21	20	18	20
139	SONAWANE BHAVESH NITIN	20	21	21	21	18	18
140	SONAWANE SOHAM PRAMOD	20	20	20	20	18	18
141	THAKARE SAKSHI DINESH	20	18	19	20	18	17
142	TIWARI MANAS SANTOSHKUMAR	21	20	17	20	19	19
143	WAGH SIDDHARTH PRAKASH	21	22	23	23	23	23
	No of Students appeared	143	143	143	143	143	143
	No of Students achieving target	144	143	143	142	129	143
	% of Students achieving target	100	100	100	99.301	90.21	100
	Covered CO's	CO1	CO2	CO3	CO4	CO5	CO6
	Co Attainment	3	3	3	3	3	3

Name & Signature of Faculty
Ms. Chaitali S. Patil

HOD
Mr. N. D. Patel

Fig.(SLA)Self Learning Assessment

2. Literature Review

S.S.V.P.S. Bapusaheb Shivajirao Deore Polytechnic , Dhule

Department of Computer Engineering

FORMATIVE ASSESSMENT OF PRACTICAL (FA-PR)

Institute Code and Name:0059
Academic Year : 2024-25
Program:CO4K
Semester: IV

Exam: Summer 2025
Course: JPR Course Code : 314317
Marks: Max- 25 Min - 10
Name of Faculty : Ms. Chaitali S. Patil

K3

Roll No.	Name of Student	Experiment/ Job /Assignment / Sheet/Activity of Project (Marks out of 50 per experiment)																				FA Marks of Practical converted According to L-A SchemeMax Marks (50)
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
		CO1	CO1	CO1	CO1	CO2	CO2	CO2	CO2	CO2	CO3	CO3	CO3	CO3	CO4	CO4	CO4	CO5	CO5	CO6	CO6	
1	AHIRE SIDDHESH SUNIL	17	19	17	17	17	17	17	17	17	17	17	19	19	17	17	17	17	17	17	17	18
2	AHIRE TANUSHREE PARAG	18	18	15	15	15	15	19	15	20	20	19	15	18	15	18	15	18	18	18	15	17
3	ANSARI MOHAMMAD SOHAIL MOHAMMAD HUSAIN	19	19	17	18	17	18	17	17	18	17	18	19	18	19	17	19	18	19	17	17	18
4	ANSARI ZEBA JAVEED IQBAL	21	21	22	21	22	22	22	22	22	21	21	21	21	21	21	21	21	21	21	21	22
5	BAISANE NARENDRA PRABHAKAR	18	19	17	18	18	18	17	17	17	17	17	19	18	19	18	17	19	16	19	20	18
6	BHAMARE PIYUSH PRASHANT	22	22	22	22	22	22	22	22	22	22	21	21	21	21	21	21	21	21	21	21	22
7	BHAMARE VAISHNAVI SUNIL	18	18	18	18	18	18	18	17	18	17	17	17	20	17	20	17	17	20	17	17	18
8	BHAVSAR SATYAM PARESH	20	19	19	19	19	18	17	19	18	19	17	17	18	17	19	17	17	17	21	21	19
9	BOHRI HATIM HUZEFA	22	22	20	21	20	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21
10	BORASE PRATHAMESH UDAY	17	17	17	17	17	17	17	17	17	17	18	18	18	18	18	18	18	18	18	18	18
11	BORSE AASHUTOSH KANHAIYALAL	20	21	19	20	20	20	19	19	18	19	19	18	18	18	18	18	18	18	18	19	19
12	BUVA SAKSHI ANANDGIR	21	19	20	19	20	19	20	19	18	22	19	18	18	21	21	21	21	21	21	21	20
13	CHAUDHARI KARTIK BHAGWAN	21	21	22	22	22	22	22	21	21	21	21	21	21	21	21	21	21	21	21	21	22
14	CHAUDHARI KHUSHI PRAVIN	17	17	17	17	17	17	17	17	20	20	20	21	21	21	21	21	21	21	21	21	20
15	CHAUDHARI KUNAL RAGHUNATH	20	19	19	19	19	19	19	18	18	18	18	18	18	18	18	18	18	18	18	18	19
16	CHAUDHARI SAKSHI JAIPRAKASH	11	12	11	12	12	11	12	12	11	12	11	11	12	11	12	12	12	12	12	12	12
17	CHAUDHARI SIDDHESH PRAKASH	18	18	17	17	17	17	17	17	17	17	17	17	17	18	18	18	18	18	18	18	18
18	CHAUDHARI YASHIKA PARESH	16	16	16	18	16	18	16	19	19	15	19	19	19	19	19	19	19	19	19	15	18

Fig.(FA-PR) Formative Assessment Of Practical

2. Literature Review

	Name of Student	Experiment/ Job /Assignment / Sheet/Activity of Project (Marks out of 50 per experiment)																				FA Marks of Practical converted According to L-A SchemeMax Marks (50)
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
		CO1	CO1	CO1	CO1	CO2	CO2	CO2	CO2	CO2	CO3	CO3	CO3	CO3	CO4	CO4	CO4	CO5	CO5	CO6	CO6	
139	SONAWANE BHAVESH NITIN	19	19	19	17	19	17	19	17	19	17	16	17	19	19	17	17	19	17	17	19	18
140	SONAWANE ROHAN PRAMOD	16	19	16	16	16	16	16	19	19	19	16	16	16	16	19	19	19	19	19	19	18
141	THAKARE SAKSHI DINESH	12	11	12	11	12	11	12	11	12	11	12	12	11	12	11	11	11	11	11	12	12
142	TIWARI MAHAK SANTOSHKUMAR	12	12	11	11	11	12	11	11	12	11	11	12	12	12	11	12	11	11	12	12	12
143	WAGH SIDDHARTH PRAKASH	22	22	21	22	21	21	21	21	22	22	21	21	21	21	21	21	22	22	22	22	22
	No of Students appered	143	143	143	143	143	143	143	143	143	143	143	143	143	143	143	143	143	143	143	143	143
	No of Students achiving target	133	133	133	133	133	133	133	133	133	133	133	133	133	133	133	133	133	133	133	133	133
	% of Students achivng target	93	93	93	93	93	93	93	93	93	93	93	93	93	93	93	93	93	93	93	93	93.00699
	Covered CO's	CO1	CO1	CO1	CO1	CO2	CO2	CO2	CO2	CO2	CO3	CO3	CO3	CO3	CO4	CO4	CO4	CO5	CO5	CO6	CO6	
	Co Attainment	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

CO1	3
CO2	3
CO3	3
CO4	3
CO5	3
CO6	3

Name & Signature of Internal Faculty

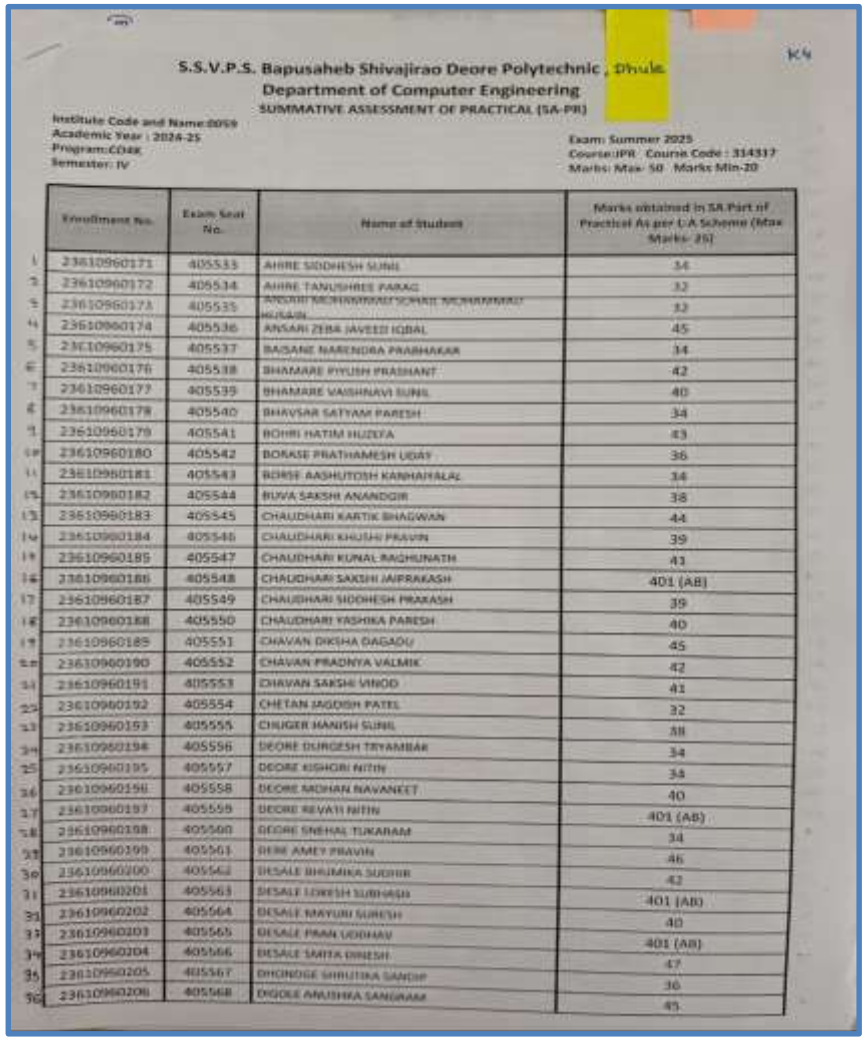
Ms. Chaitali S. Patil

HOD

Mr. N. D. Patel.

Fig.(FA-PR) Formative Assessment Of Practical

2. Literature Review



S.S.V.P.S. Bapusaheb Shivajirao Deore Polytechnic , Dhule
Department of Computer Engineering
SUMMATIVE ASSESSMENT OF PRACTICAL (SA-PR)

Institute Code and Name:0059
Academic Year : 2024-25
Program:COAX
Semester: IV

Exam: Summer 2025
Course:IPR Course Code : 314317
Marks: Max- 50 Marks Min-20

	Enrollment No.	Exam Seat No.	Name of Student	Marks obtained in SA Part of Practical As per CA Scheme (Max Marks- 25)
1	23610960171	405533	AHIRE SIDDHESH SUNIL	34
2	23610960172	405534	AHIRE TANUSHREE PARAG	32
3	23610960173	405535	ANSARI MOHAMMAD SOHAIL MOHAMMAD HUSAIN	32
4	23610960174	405536	ANSARI ZEBA JAVEED IQBAL	45
5	23610960175	405537	BAGANE NARENDRA PRABHAKAR	34
6	23610960176	405538	BHAMARE PIYUSH PRASHANT	42
7	23610960177	405539	BHAMARE VAISHNAVI SUNIL	40
8	23610960178	405540	BHAVSAR SATYAM PARESH	34
9	23610960179	405541	BOHRI HATIM HUZEFA	43
10	23610960180	405542	BORASE PRATHAMESH UDAY	36
11	23610960181	405543	BORSE AASHUTOSH KANHAIYALAL	34
12	23610960182	405544	BUVA SAKSHI ANANDGIR	38
13	23610960183	405545	CHAUDHARI KARTIK BHAGWAN	44
14	23610960184	405546	CHAUDHARI KHUSHI PRAVIN	39
15	23610960185	405547	CHAUDHARI KUNAL RAGHUNATH	41
16	23610960186	405548	CHAUDHARI SAKSHI JAIPRAKASH	401 (AB)
17	23610960187	405549	CHAUDHARI SIDDHESH PRAKASH	39
18	23610960188	405550	CHAUDHARI YASHIKA PARESH	40
19	23610960189	405551	CHAVAN DIKSHA DAGADU	45
20	23610960190	405552	CHAVAN PRADNYA VALMIK	42
21	23610960191	405553	CHAVAN SAKSHI VINOD	41
22	23610960192	405554	CHETAN JAGDISH PATEL	32
23	23610960193	405555	CHUGER MANISH SUNIL	38
24	23610960194	405556	DEORE DURGESH TRYAMBAK	34
25	23610960195	405557	DEORE KISHORI NITIN	34
26	23610960196	405558	DEORE MOHAN NAVANEET	40
27	23610960197	405559	DEORE REWATI NITIN	401 (AB)
28	23610960198	405560	DEORE SNEHAL TUKARAM	34
29	23610960199	405561	DERE AMEY PRAVIN	46
30	23610960200	405562	DESALE BHUMIKA SUDHIR	42
31	23610960201	405563	DESALE LOKESH SUBHASH	401 (AB)
32	23610960202	405564	DESALE MAYURI SURESH	40
33	23610960203	405565	DESALE PRAN UDDHAV	401 (AB)
34	23610960204	405566	DESALE SMITA DINESH	47
35	23610960205	405567	DHONDGE SHRUTIKA SANDIP	36
36	23610960206	405568	DIGOLE ANUSHKA SANGRAM	45

Fig. (SAPR)-Summative Assessment Of Practical

2. Literature Review

	Enrollment No.	Exam Seat No.	Name of Student	Marks obtained in SA Part of Practical As per L-A Scheme (Max Marks- 25)
121	23610960300	405653	VYAVAHARE HIMANSHU GHANSHYAM	36
122	23610960301	405654	WAGH DHIRAJ PRAVIN	34
123	23610960302	405655	WAGH KUNAL SAMBHAJI	44
124	23610960303	405656	WANI DIPESHA AJAY	34
125	23610960304	405657	WASWANI DIA DINESH	47
126	23610960305	405658	YOGESHWARI SUDHIR CHAUDHARI	47
127	24610960301	405659	BHAMARE DEVANGI PRAVIN	40
128	24610960302	405660	CHAVAN MANASI DEVIDAS	40
129	24610960303	405661	JADHAV YASH JAYSING	38
130	24610960304	405662	JOSHI PRATHAMESH NISHIKANT	32
131	24610960305	405663	KHAIRNAR ROHIT DILIP	41
132	24610960306	405664	MARATHE LATIKESH SUBHASH	44
133	24610960307	405665	PADMAR PRAMOD DNYANESHWAR	34
134	24610960308	405666	PADVI MITESH AJABSING	32
135	24610960309	405667	PATHAN SOHELKHAN MUSHIRKHAN	45
136	24610960311	405668	PATIL KALYANI MAHESH	34
137	24610960312	405669	PATIL PRITI TUKARAM	41
138	24610960313	405670	SHIRSATH DARSHAN ATUL	34
139	24610960314	405671	SONAWANE BHAVESH NITIN	32
140	24610960315	405672	SONAWANE ROHAN PRAMOD	33
141	24610960316	405673	THAKARE SAKSHI DINESH	401 (AB)
142	24610960317	405674	TIWARI MAHAK SANTOSHKUMAR	401 (AB)
143	24610960318	405675	WAGH SIDDHARTH PRAKASH	45

No of Students appered	132
No of Students achieving target	132
% of Students achieving target	100
Covered CO's	CO1,CO2,CO3,CO4,CO5,CO6
Co Attainment	3

Name and Signature of Faculty
Ms. Chaitali S. Patil

HOD
Mr. N. D. Patel.

Fig. (SA-PR)-Summative Assessment Of Practical

3. Scope of the project

The scope of this project is defined within the academic framework of institutes affiliated under MSBTE. The system is intended for implementation at S.S.V.P.S's B. S. Deore Polytechnic, Dhule, specifically within the Computer Engineering Department. It supports internal academic evaluation processes followed under various curricula such as the K Scheme and I Scheme. The project focuses on subject-wise performance analysis for the current batch of students and provides structured support to faculty members and administrative authorities. Its scope includes attainment calculation, rubrics-based classification, gap identification, and generation of analytical reports for internal academic review. The project is limited to institutional-level evaluation support and does not extend to external examination or result processing systems.

4. Methodology

- <u>Technologies used:-</u>

1. Django(Python):-Used for backend development. It processes the uploaded SRS file, performs validation, applies NLP techniques, and handles communication with the database.

   - https://www.python.org/doc/

2. Sqlite3:- Used as the database to store user and admin data.

   - https://www.sqlite3.com/docs/

3. Frontend:- Django templates (HTML/CSS/JS), Bootstrap, jQuery, DataTables . JSON Fields: Used extensively to store flexible schemas (e.g., subjects_data, students_data, student_marks, configs).

4. Methodology

- The system is a web-based application designed to automate CO attainment calculation for SSVPS College.
- It allows faculty to enter and manage student assessment data in a structured and accurate way.
- The system automatically calculates CO attainment percentages after uploading student scores.
- It helps in monitoring student performance and identifying learning gaps early.
- Graphical visualizations (like bar charts) support analysis and reduce manual effort with real-time monitoring.

5. Details of designs, working & processes

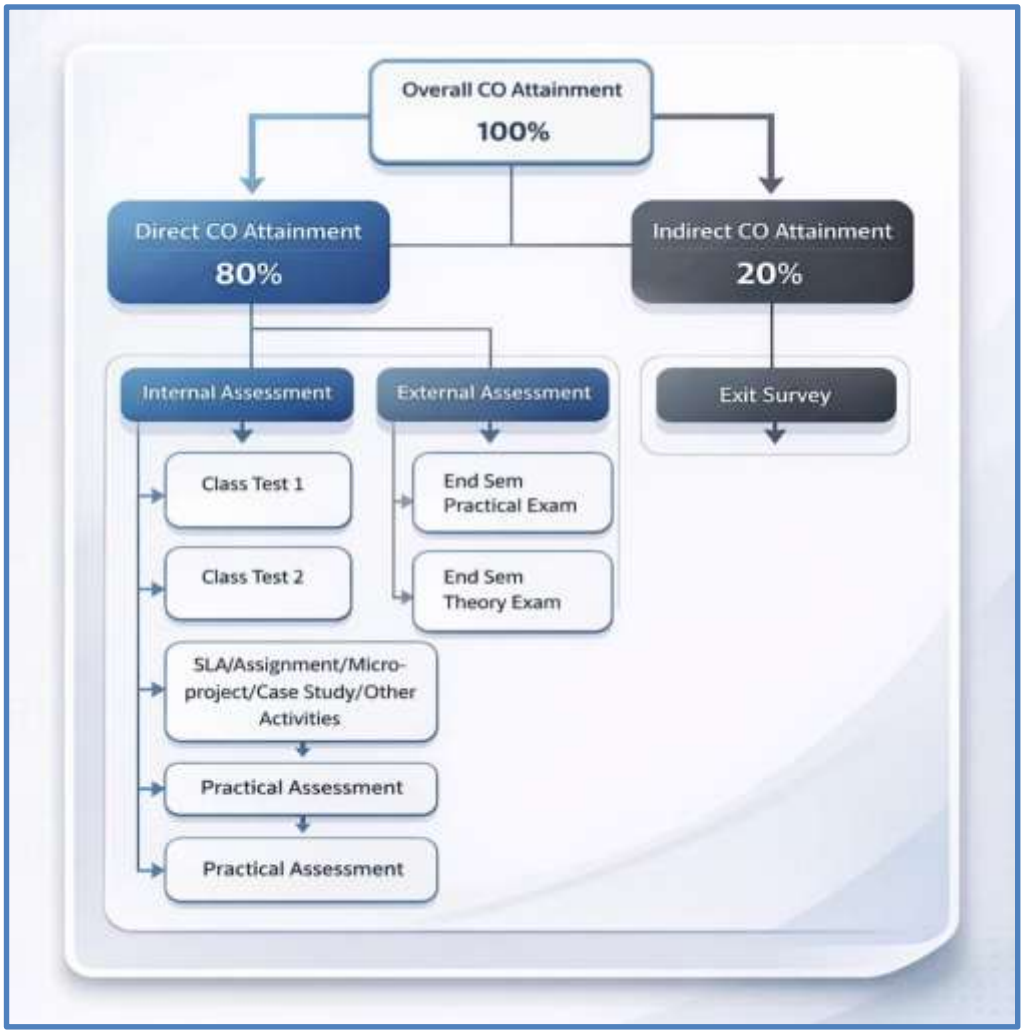
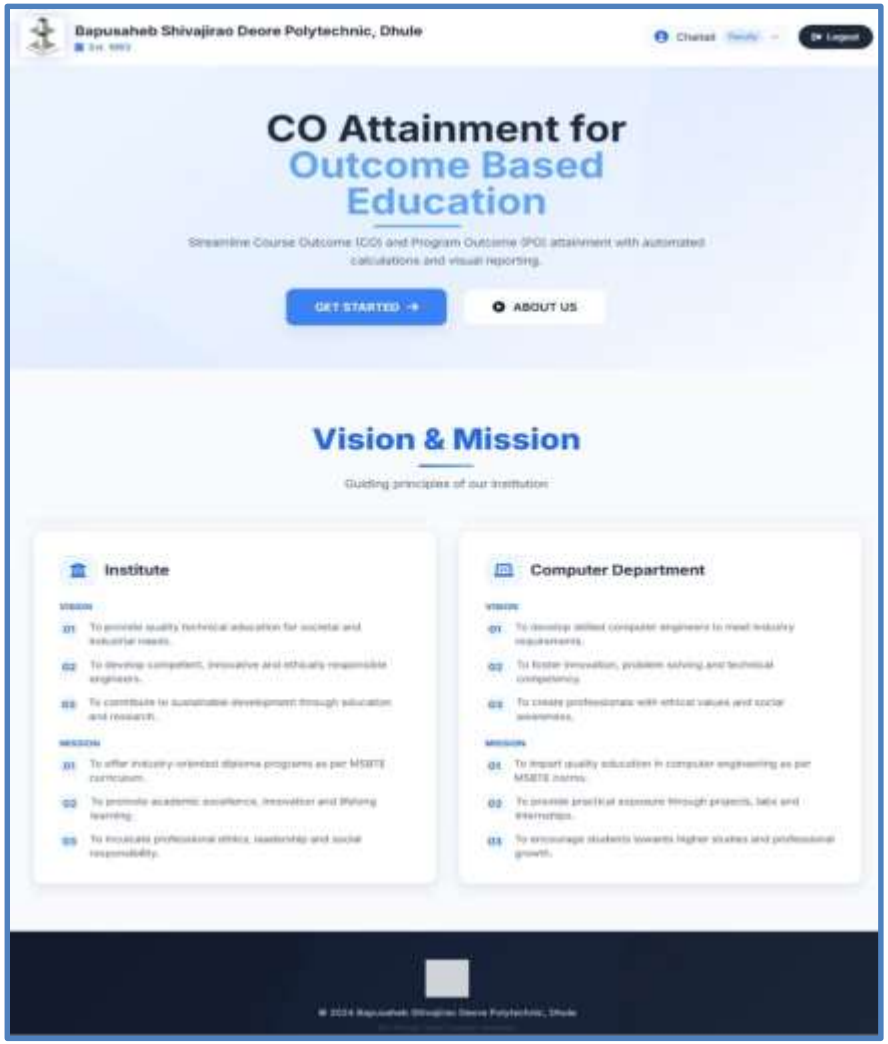


Fig.CO-Attainment Process

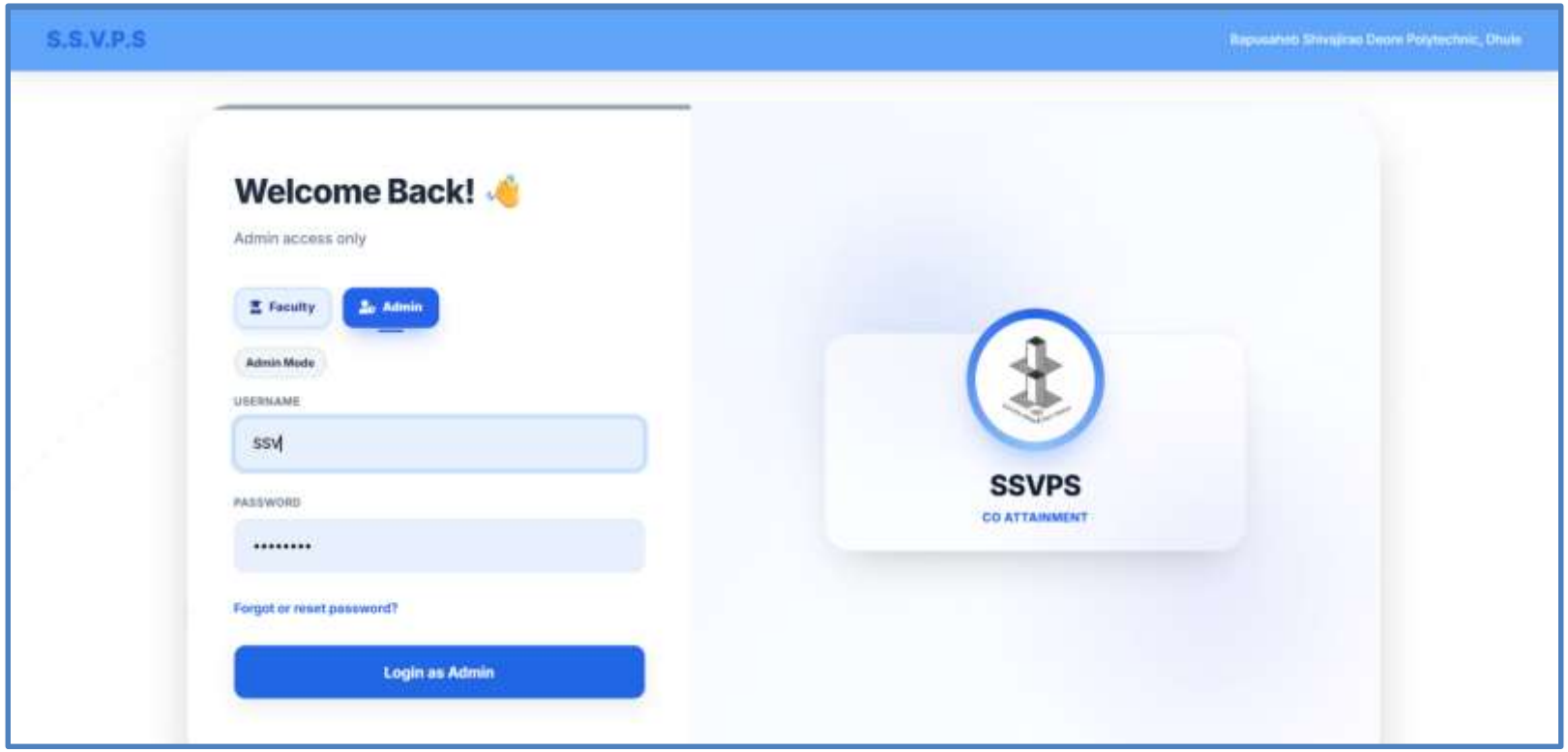
6. Results and Applications



Home Page :-

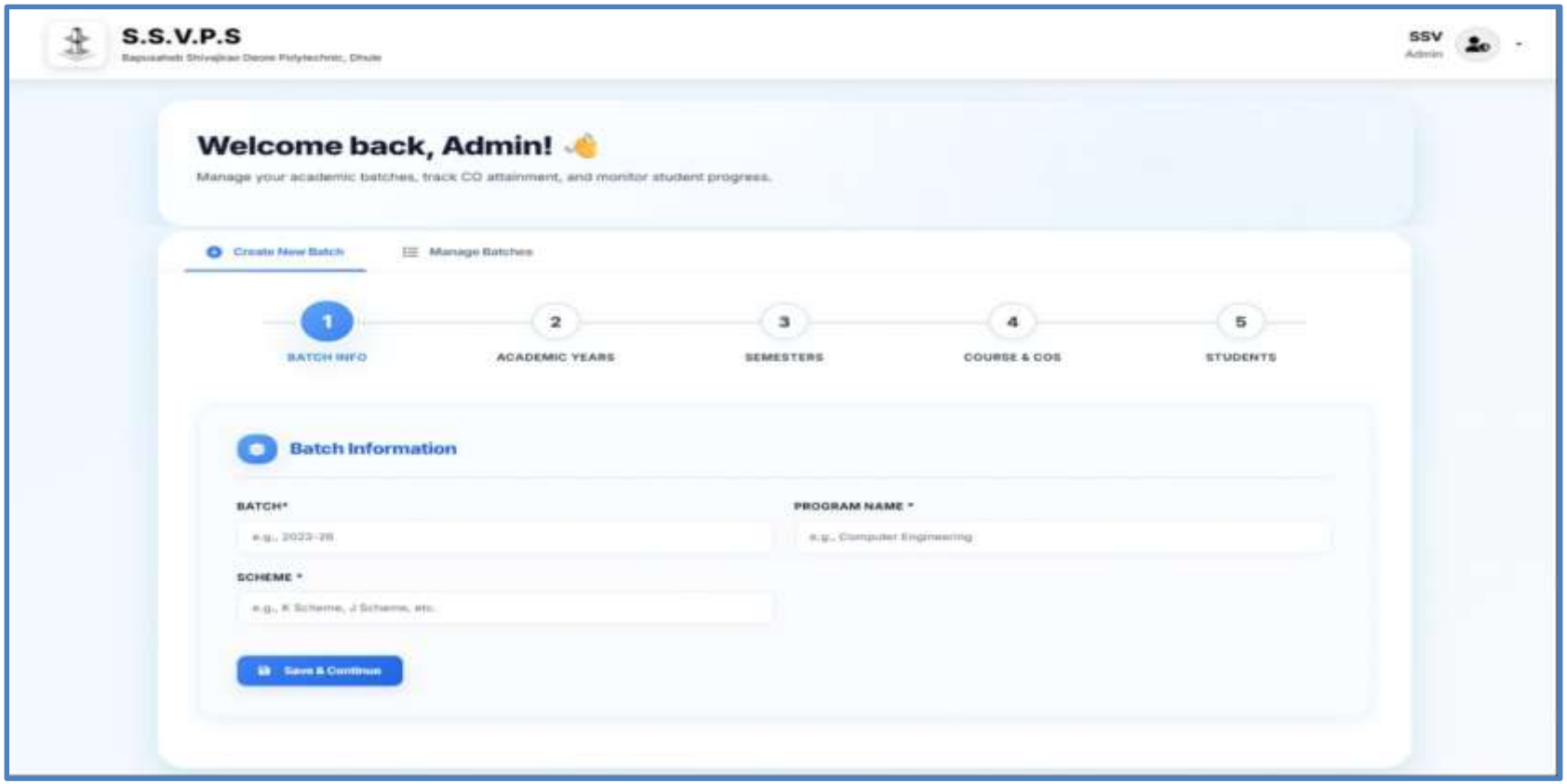
This is the home page of the CO Attainment System. It provides an overview of the system and its purpose, which is to calculate Course Outcome attainment automatically. The page also displays the institute and department vision and mission to highlight the academic objectives of the institution.

6. Results and Applications



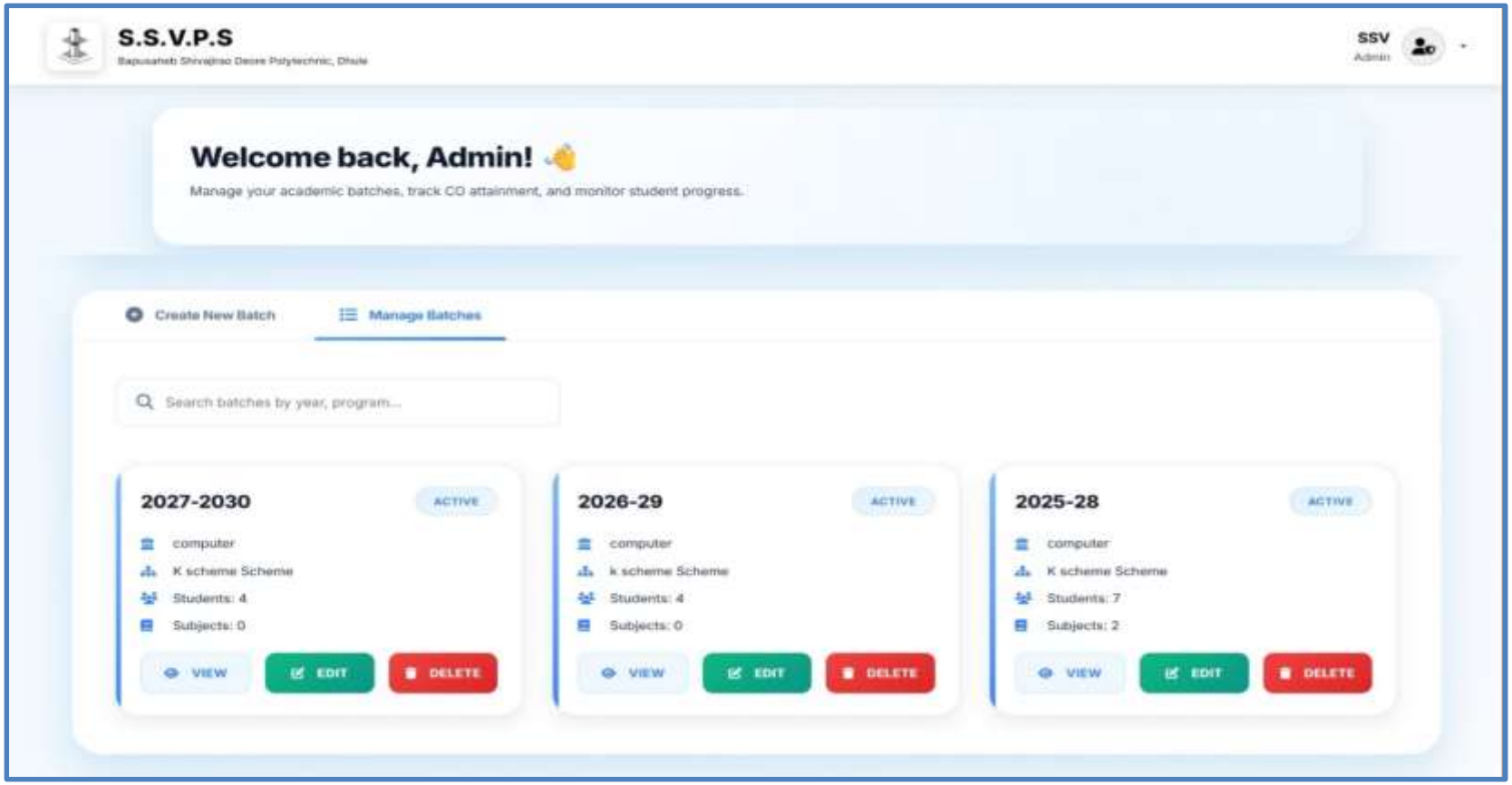
Login Page:- This page provides secure access to the system through the Admin login interface. The administrator enters valid login credentials to access the system. After successful login, the admin can manage academic batches, courses, and CO attainment related data within the system.

6. Results and Applications



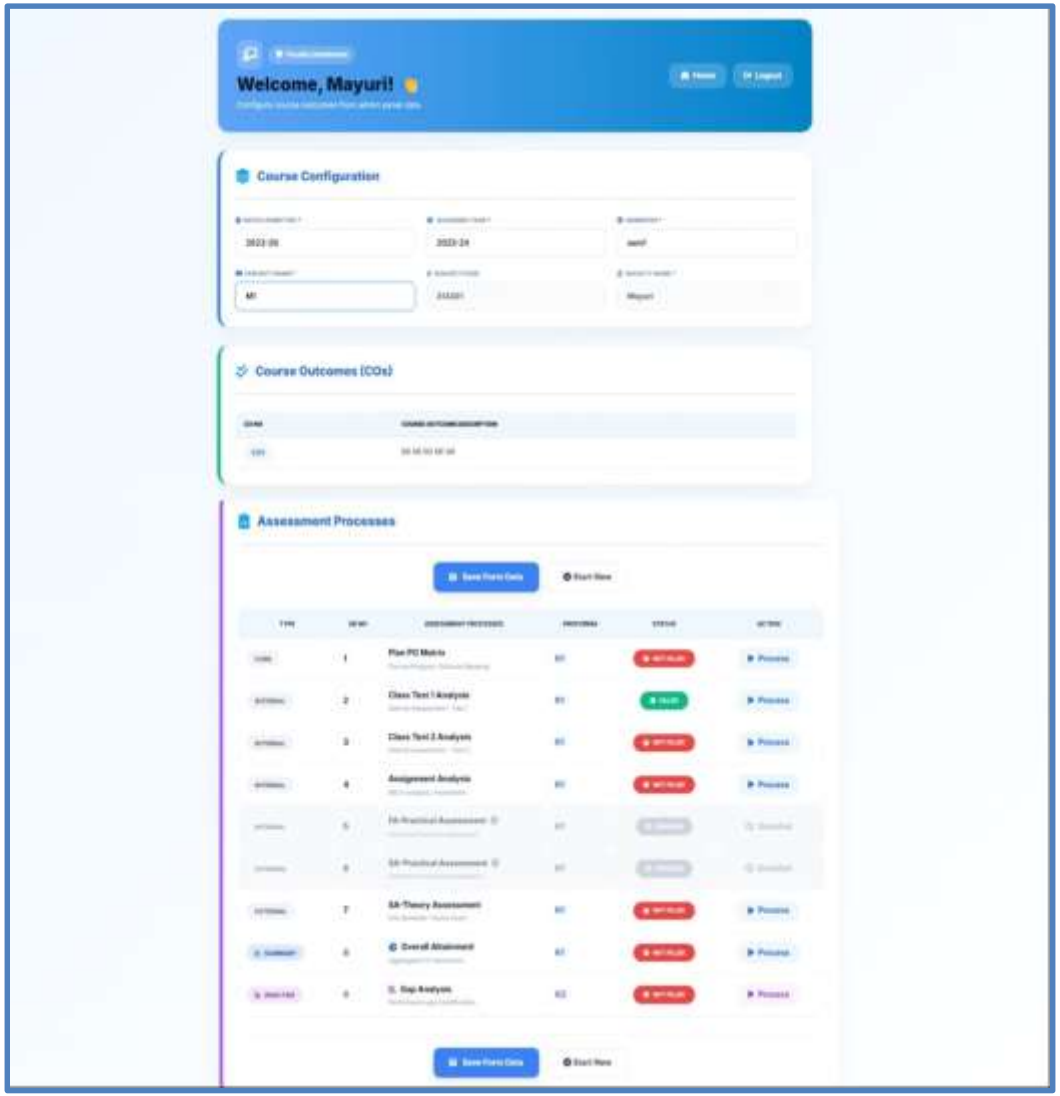
<u>Admin Dashboard (Create Batch):-</u> This page allows the admin to create and manage academic batches. The admin enters important details . These details help organize student and course information for CO attainment calculation.

6. Results and Applications



Manage Batches:- This page displays all the created batches in the system. The admin manage batch information and check the number of students and subjects in each batch. It helps in managing academic records efficiently.

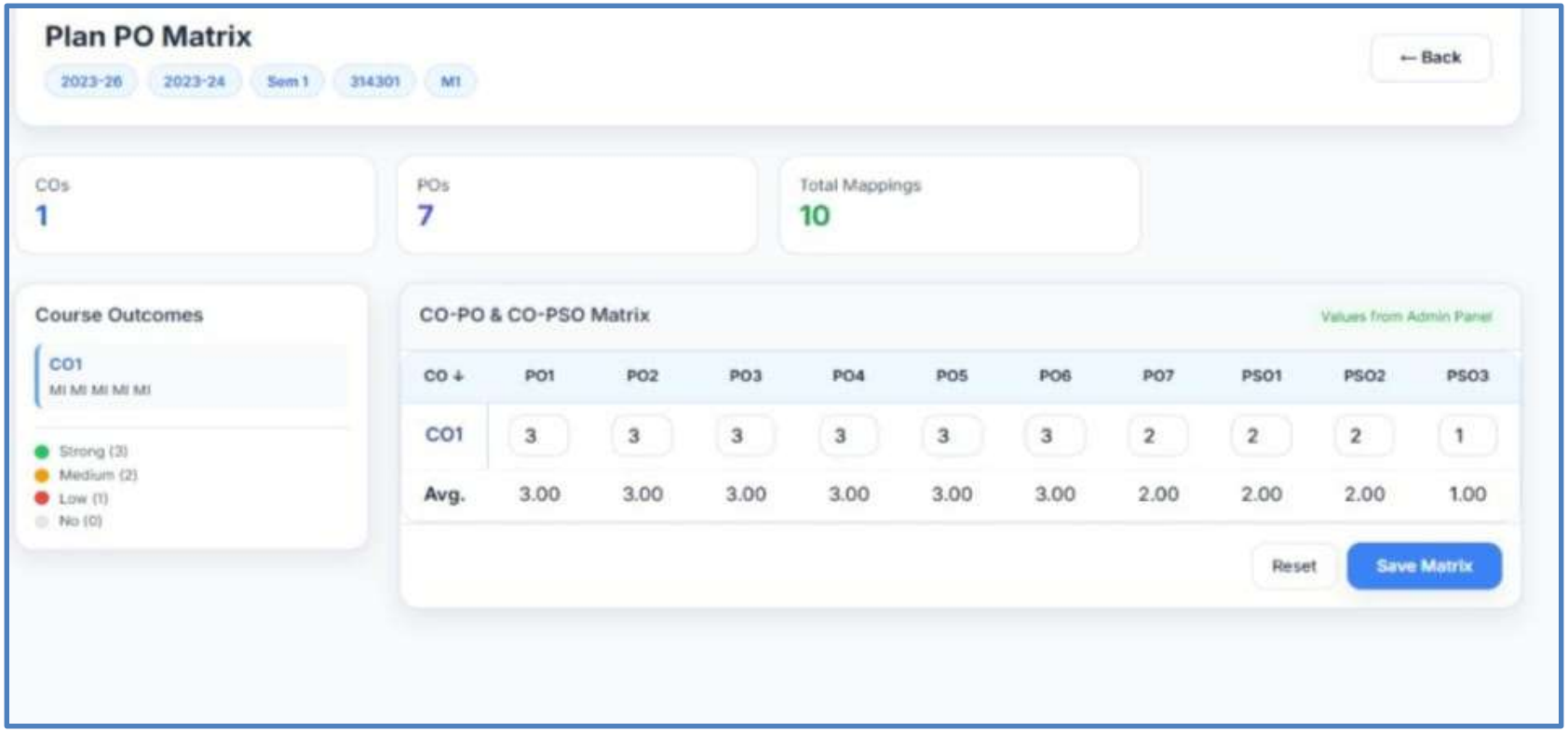
6. Results and Applications



Course Configuration:-

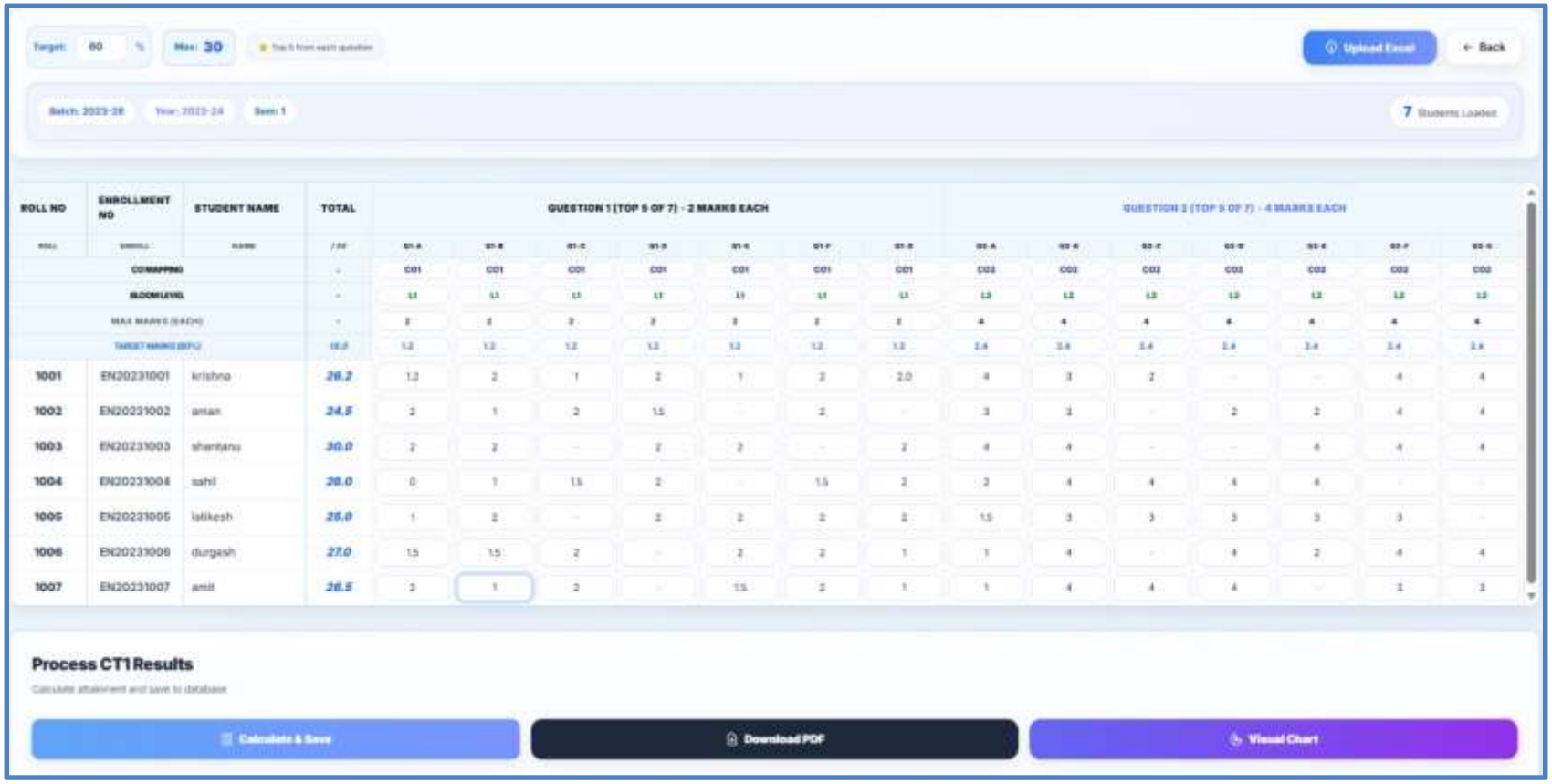
This interface shows the Teacher Module, where teachers configure course details and define Course Outcomes (COs). It also provides different assessment processes such as PO Matrix mapping, class test analysis, assignment evaluation, and overall attainment calculation.

6. Results and Applications



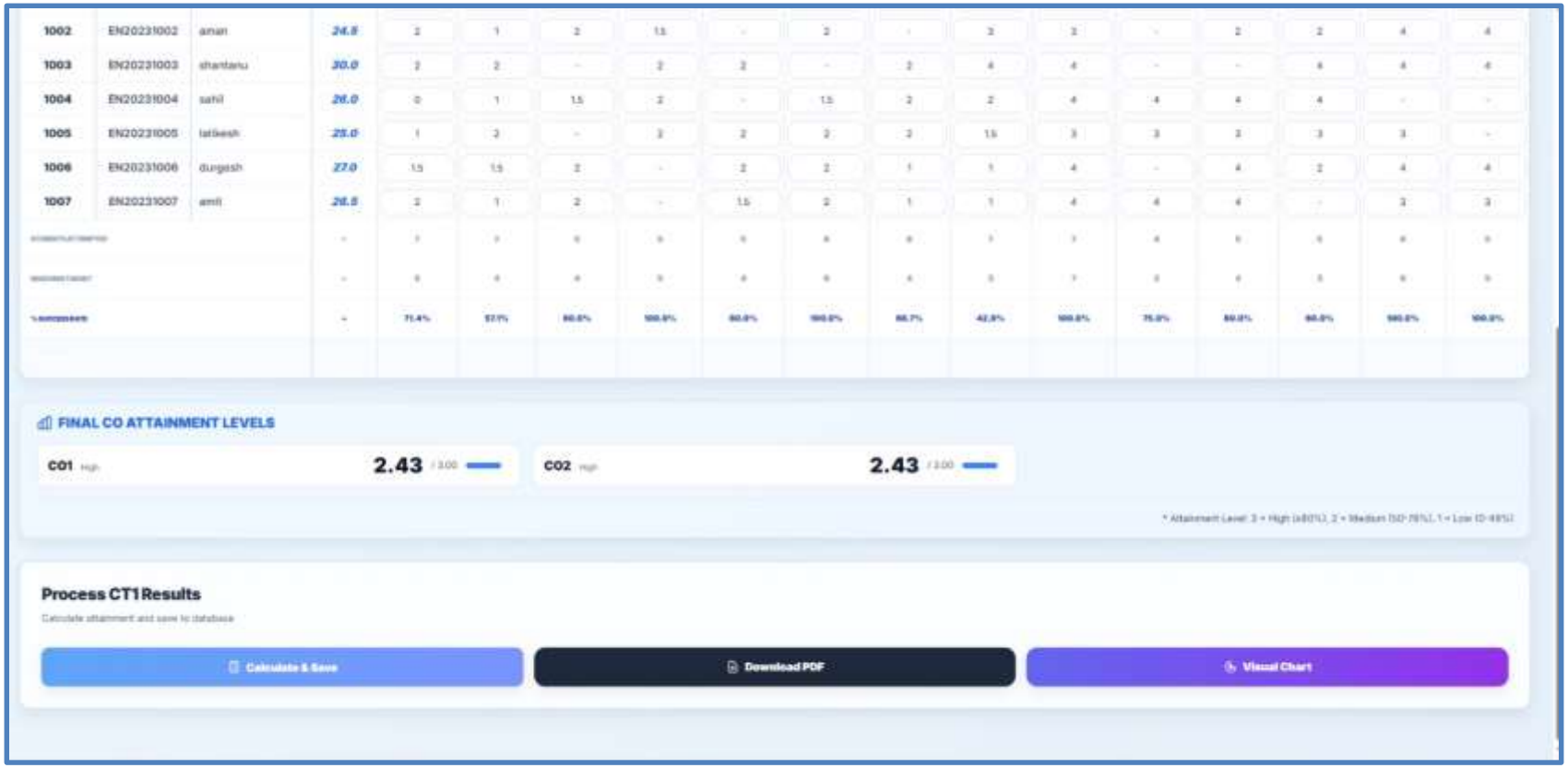
<u>Plan PO Matrix Interface:-</u>This interface shows the CO-PO & CO-PSO Matrix, where Course Outcomes are mapped with Program Outcomes and PSOs. The course and subject details configured in the Admin Panel are automatically reflected here, enabling teachers to define and save the PO mapping for assessment.

6. Results and Applications



Class Test Interface:- This interface is used for Class Test assessment, where teachers can enter student marks based on CO mapping. The system then calculates the results and provides attainment analysis, reports, and visual charts.

6. Results and Applications



This interface shows the final CO attainment levels calculated from the student marks entered in the assessment. It displays the attainment value for each CO and helps in evaluating the overall performance of students.

7. Conclusion and future scope

Conclusion:-

The proposed web-based CO attainment system provides a centralized and automated solution for monitoring Faculty Members performance. By allowing faculty to enter, manage and calculate CO data efficiently, the system ensures accurate and timely evaluation of learning outcomes.

The system also offers graphical visualization for target attainment and actual attainment, helping faculty identify performance gaps to improve Faculty Members performance by manually setting their action plan. Overall, the project saves time, enhances data-driven decision-making, and supports academic contributing to better teaching strategies effectiveness.

7. Conclusion and future scope

Future Scope :-

In future, the system can be extended to include Program Outcome (PO) and Program Specific Outcome (PSO) attainment for comprehensive academic evaluation. Advanced dashboards can be implemented to provide semester-wise and subject-wise performance trends using enhanced visual analytics. The system can also be enhanced to support student-wise performance analysis for personalized academic monitoring. Additionally, automated action plan recommendations based on identified gaps can be integrated to assist faculty in improving learning outcomes.

8. References and Bibliography

1. Bhivrabai Sawant Polytechnic.(**NBA-** National Board Of Acceredation)

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2.https://www.irjweb.com/CO%20AND%20PO%20ATTAINMENT%20CALCULATOR.pdf